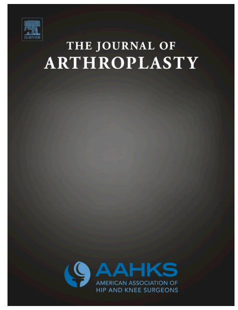


# Effectiveness of Telerehabilitation After Total Knee Arthroplasty: An Umbrella Review

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# Effectiveness of Telerehabilitation After Total Knee Arthroplasty: An Umbrella Review


## ABSTRACT

**Background:** The use of telerehabilitation is increasing worldwide among patients who have undergone total knee arthroplasty (TKA). However, its efficacy and credibility in clinical settings remain unclear.

**Objectives:** An umbrella review (CRD420251070340) of the available scientific evidence was performed to determine whether telerehabilitation is an effective alternative to conventional rehabilitation among patients after TKA.

**Methods:** We searched multiple databases through July 30, 2025, for relevant systematic reviews or meta-analyses. Effect sizes were recalculated as standardized mean differences (SMDs) with 95% confidence intervals (CIs). The initial search retrieved 1,192 articles.

**Results:** The trials showed a significant effect of telerehabilitation interventions on patients undergoing TKA (SMD = -0.06, 95% CI [-0.16 to 0.03], $P < 0.001$, $I^2 = 69.9\%$). The bias estimate was -0.423 (standard error (SE) = 0.809, $P = 0.604$). In subgroup analyses, telerehabilitation interventions all had a significant effect on patients who underwent TKA (SMD = -0.14, 95% CI [-0.28 to 0.01], $P < 0.001$, $I^2 = 73.9\%$; SMD = 0.02, 95% CI [-0.09 to 0.14], $P = 0.005$, $I^2 = 47.7\%$; SMD = -0.15, 95% CI [-8.53 to 8.23], $P < 0.001$, $I^2 = 95.9\%$, respectively). Range of motion (passive flexion) (SMD = 0.26, 95% CI [-0.06 to 0.58], $P = 0.041$, $I^2 = 59.8\%$), Western Ontario and McMaster Universities Osteoarthritis Index (SMD = -0.40, 95% CI [-0.75 to -0.05], $P = 0.003$, $I^2 = 69.9\%$), and Knee Injury and Osteoarthritis Outcome Score (SMD = -0.57, 95% CI [-3.52 to 2.39], $P < 0.001$, $I^2 = 85\%$) all showed improvement over traditional face-to-face rehabilitation among patients who underwent TKA.

**Conclusions:** This body of evidence tends to support telerehabilitation compared with traditional face-to-face rehabilitation, as indicated by better range of motion (passive flexion), the Western Ontario and McMaster Universities Osteoarthritis Index, and the

Knee injury and Osteoarthritis Outcome Score among patients who underwent TKA.

**Keywords:** total knee arthroplasty; telerehabilitation; umbrella review; meta-analysis; systematic review; patients

## Introduction

Knee osteoarthritis is increasingly recognized as a worldwide health problem, with a global estimated incidence of 4,307.4 per 100,000 people [1]. Total knee arthroplasty (TKA) is the gold-standard treatment for end-stage knee osteoarthritis when conservative treatment fails to relieve symptoms [2]. Although this operation relieves pain and improves functions, up to 30% of patients are, to some extent, dissatisfied with the result, and re-operation likely would result in a high economic burden on individuals and society [3, 4]. Risk factors include lack of social exposure to others, lack of participation in social activities, pain catastrophizing, low self-efficacy, and negative expectations regarding postoperative kneeling or psychological well-being [5].

Wound care and postoperative rehabilitation are essential components of treatment [6]. They can relieve pain; enhance muscle strength; prevent complications such as deep vein thrombosis, joint stiffness, and wound infection; promote joint function recovery; and improve independent living ability and quality of life [4]. However, owing to transportation constraints, appointment scheduling conflicts [7], financial constraints [8], and staff shortages in the healthcare sector [9,10], in-person participation can be challenging, particularly during pandemic situations that involve social distancing measures. Consequently, patient enrollment rates may be low and dropout rates may be high, thus preventing patients from successfully integrating into their community and living independently [11].

Telerehabilitation can increase accessibility of care and reduce the burden of time and financial costs of traveling to a clinic [12]. It refers to remote medical treatment through communication and information technologies such as the internet, smartphones, virtual reality, video phones, and wearable electronic devices, providing patients who are undergoing timely rehabilitation guidance, remote supervision, and no travel distance. Compared with conventional rehabilitation, telerehabilitation saves time and cost, enables patients to obtain more convenient home rehabilitation services, enhances enthusiasm for rehabilitation, helps ensure continuous nursing care, and improves quality of life [4, 13].

Systematic reviews of the effectiveness of telerehabilitation for patients undergoing TKA have gradually increased in recent years and showed positive results. For instance, evidence from a systematic review [14] showed improvements in stiffness scores on the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) and the six-minute walk test following telerehabilitation relative to in-person care. However, the quality of the studies, reported results, and conclusions appeared to differ, making clinical decision-making challenging, and there are no universally accepted guidelines for telerehabilitation after TKA [15,16]. Thus, the extent and certainty of the evidence on the effectiveness of telerehabilitation in improving motor function, pain, acceptability, and treatment attendance and adherence after TKA are unknown. In addition, different statistical methods and approaches to evaluating the certainty of the evidence might have been used in prior meta-analyses.

To address these gaps, we conducted an umbrella review of comprehensive meta-analyses to combine the evidence on various outcomes after TKA, comprehensively examine available information on telerehabilitation, and compare the results of published systematic reviews to provide a broader understanding of telerehabilitation approaches among patients who have undergone TKA [17]. Umbrella reviews extend systematic reviews by identifying, synthesizing, and evaluating evidence from multiple systematic reviews and meta-analyses on a common topic [18]. The current umbrella review had two goals: (a) to determine the influence of telerehabilitation (mobile devices [19]; video conferencing [20]; fixed equipment like augmented reality (AR) and virtual reality (VR) [21]; wearable devices such as sensors [22]; and so on) compared with traditional face-to-face rehabilitation; and (b) to examine the effectiveness of telerehabilitation on pain relief, range of motion (ROM), WOMAC, Knee Injury and Osteoarthritis Outcome Score (KOOS), satisfaction, acceptability, attendance, adherence, and so on.

## Method

This review complied with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses reporting checklist, as detailed in Appendix 1. The protocol was registered with the International Prospective Register of Systematic Reviews (registration number: CRD420251070340).

### Eligibility criteria

The populations, interventions, comparisons, outcomes, and study types framework was used to develop the following inclusion criteria:

1. Population: Patients aged 18 years or older who have been clearly diagnosed with knee osteoarthritis and have undergone TKA in any country or region.
2. Intervention: Telerehabilitation involving at least one technological aid (e.g., wearable devices, mobile applications, video guidance, or remote monitoring) and specific content: home rehabilitation training programs, data tracking, remote medical consultation, or rehabilitation.
3. Comparisons: Traditional face-to-face rehabilitation (such as outpatient physical therapy, self-exercise at home without remote guidance) or other non-remote intervention methods.
4. Outcome: Primary outcome of knee joint function recovery (e.g., WOMAC, ROM), pain relief (Visual Analog Scale [VAS] score), and secondary outcomes of patient satisfaction, quality of life (36-Item Short Form Health Survey), and cost-effectiveness of rehabilitation.
5. Study design: Systematic reviews and meta-analyses of randomized

controlled trials (RCTs), excluding network meta-analyses.

We excluded (a) hybrid interventions (e.g., telerehabilitation component less than 50%); (b) non-TKA patients (e.g., hip arthroplasty, and nonsurgical rehabilitation patients); (c) major comorbidities (e.g., neurological disorders, and poly-articular pathologies); and (d) animal research and conference abstracts. In case of overlap in RCTs of included studies, we selected studies with the most RCTs to be included in the overall review to avoid double-counting [23].

All search results were imported into EndNote version X9 (Thomson Research Soft, Stamford, Connecticut, USA) for deduplication. Titles and abstracts were independently screened for relevance by two reviewers (Hp. X and Zx, G), and the full texts of potentially eligible articles were assessed. Any disagreements were resolved through discussion with a third reviewer (Xy, Z). Meta-analyses were assessed by full-text reading by two reviewers before inclusion in the umbrella review.

**Search strategy**

We searched PubMed, MEDLINE, Web of Science, Cochrane Library, CINAHL, Science Direct, Embase, and Scopus from inception to July 30, 2025. Our search strategy adopted a combination of keywords related to telerehabilitation and TKA to expand the search scope and derive comprehensive evidence on the topic ( Appendix 2).

**Data extraction**

The web-based literature review tool Covidence (http://www.covidence.org) was used to facilitate the systematic review process. After identifying all relevant articles and removing duplicates, two reviewers (Hp. X and Zx. G.) screened the titles, abstracts, and full-text articles together, with verification by another reviewer (Xy, Z). Any discrepancies were resolved through consensus. The following components for each article were extracted by two reviewers, stored, and synthesized in Microsoft Excel: (a) author and publication year; (b) study country; (c) total reviews included; (d) number of original studies; (e) population (sample size, eligibility criteria, and exclusion criteria); (g) interventions (setting, categorization, and duration); (f) outcomes measured; (h) key results; (i) meta-analysis (yes or no); and (j) quality assessment tools (see Table 1).

**Quality assessment and certainty of evidence**

The quality of the meta-analyses was evaluated using the 16-item Measurement Tool to Assess Systematic Reviews 2 (AMSTAR 2) [24]. There were two researchers (Hp. X and Zx, G.) who independently rated the results as high, moderate, low, or critically low. Reviews with no or only one noncritical weakness were classified as having high confidence; studies with more than one noncritical weakness were classified as moderate; studies with one critical flaw (with or without a noncritical weakness) were rated as low; and studies with multiple critical flaws (with or without noncritical flaws) were rated as critically low.

The degree of certainty of the results was evaluated using the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) method [25]. The degree of evidence depended on the risk of bias, inconsistency, imprecision, indirectness, and potential for publication bias. We graded the certainty of evidence (high, moderate, low, or very low) using the GRADE criteria.

**Overlap in individual studies in the included systematic reviews**

The degree of overlap among individual studies included in systematic reviews captured in the current umbrella review was assessed using the corrected covered area (CCA) method [26]. A CCA of 100% would indicate that all reviews contained the same individual studies, whereas a CCA of 0% would indicate that each review included unique individual studies. A CCA of less than 5% suggests a slight overlap, 6 to 10% a moderate overlap, 11 to 15% a high overlap, and more than 15% a very high overlap [26].

**Characteristics of Systematic Reviews**

Initially, 1,192 articles were retrieved. After removing 168 duplicate articles and excluding 483 non-meta-analysis articles based on their titles and abstracts, the full texts of the remaining 496 articles were read. We excluded 465 articles that did not meet the inclusion criteria, leaving 31 meta-analyses. No additional search results met the inclusion criteria. After extracting original meta-analysis data, 17 studies were excluded based on irrelevant patient populations, irrelevant interventions, or not RCTs. Thus, 14 studies [19, 20, 32-43] were included in the umbrella review for recalculation. A list of excluded studies and the reasons for exclusion are provided in Appendix 3. The screening process is illustrated in Figure 1.

An overview of all included systematic reviews is presented in Table 1. There were 14 systematic reviews included that did [19, 20, 32, 40-42] or did not [22, 33-39] involve meta-analysis, with 10 (71.4%) published in the last three years (2023 to 2025). Included studies were from China (n = six, 42, 9%), the United States (n = three, 21.4%), Greece (n = one, 7.1%), Turkey (n = one, 7.1%), the United Kingdom (n = one, 7.1%), Iran (n = one, 7.1%), and India (n = one, 7.1%). The number of individuals included in each review ranged from six to 584, the number of RCTs ranged from six to 31, the sample size ranged from 142 to 4,666, and the average age of participants ranged from 60 to 73 years. The intervention group received telerehabilitation, either in isolation or combined with classical rehabilitation. Telerehabilitation was considered any technology (mobile devices, video conferencing, augmented reality or VR, wearable devices such as sensors, and so on) that allowed for the monitoring or execution of rehabilitation or therapy. The control group received routine outpatient physical therapy or home visits. The duration ranged from five days to five years. Overall, 105 outcomes were collected, including 54 physical function and activity, 13 pain assessments, six quality of life, five compliance, five satisfaction, four cost, four adverse events, and 14 other measures. The key outcomes included ROM, WOMAC, KOOS, VAS, and 36-Item Short Form Health Survey, EuroQol 5-Dimensions, muscle strength, cost, and satisfaction. The interpretations of the scales and tests are provided in Appendix 4.

**AMSTAR assessment and GRADE classification**

The methodological quality of the included reviews was assessed by AMSTAR 2, which contained 16 items for scoring. Among the 14 systematic reviews, four were rated as high quality, three as moderate quality, five as low quality, and two as very low quality. The details of the quality analysis are shown in Appendix 5.

The quality of evidence for the 32 health-related outcomes was assessed using the GRADE system; 21 outcomes were rated as moderate, eight outcomes were rated as low, and three outcomes were rated as very low. The detailed evidence quality assessment of the 32 outcomes is shown in Appendix 6.

**Overlap in individual studies in systematic reviews**

The 14 systematic reviews featured 245 individual studies that met the inclusion criteria for this umbrella review. Among them, we identified 193 unique individual studies cited across 14 reviews. We calculated the CCA using the following formula: $\frac{N-r}{r \times c-r}$, with N indicating the total number of citations included in all reviews (245), r indicating the number of rows (i.e., included reviews, 193), and c indicating the number of columns (i.e., unique publications referenced). The CCA was 2.07%, which indicates a slight overlap. However, certain systematic reviews failed to specify the year and citation of the analysis paper, which may result in a lower CCA value. The full citation matrix is presented in Appendix 7.

**Data Synthesis and Analysis**

The same inclusion criteria for the review were employed, but two criteria were added: (a) the results section contained detailed information on comparative statistical data (e.g., means, standard deviations, and 95% confidence intervals [CIs]) of main variables, and (b) data for the analyzed variables were represented in at least two studies. Summary statistics in the form of forest plots were calculated, consisting of a weighted compilation of all standardized mean differences (SMDs) and corresponding 95% CIs reported by each study and providing an indication of heterogeneity among studies.

For each association in all included meta-analyses, we recalculated the effect size as mean differences (MDs) or SMDs, corresponding 95% CIs, and heterogeneity ($I^2$) for each association using the DerSimonian and Laird random-effects model [27]. $I^2$ indicates the heterogeneity between two studies, with $I^2$ less than 50% considered small, 50 to 75% considered moderate, and $I^2$ more than 75% considered significant [28]. Statistical significance was defined as $P < 0.05$ in two-sided tests. Evidence of a small study effect was assessed using the Egger regression asymmetry test, with $P < 0.05$ demonstrating bias due to small study effects [29].

Sensitivity and subgroup analyses were performed for significant associations using the random-effects model by performing secondary calculations, excluding studies with a high risk of bias and small sample sizes (25th percentile) [30, 31]. Statistical analyses were performed using R software (version 4.4.0, R Foundation for Statistical Computing, Vienna, Austria) and the related package metafor (V4.4-0).

## Results

### Outcome Characteristics in meta-analyses

There were six meta-analyses included in this umbrella review that featured 32 effectiveness outcomes [19, 20, 32, 40-42] (Table 2), such as ROM, WOMAC, KOOS, VAS, timed up-and-go test, muscle strength, stiffness, and function.

Liu et al. [32] found a fixed equipment-assisted telerehabilitation group exhibited superior outcomes compared to the traditional face-to-face rehabilitation group in terms of pain (SMD = -0.15, 95% CI [-0.27 to -0.04], $P = 0.010$), ROM (passive flexion; MD = 2.60, 95% CI [0.77 to 4.44], $P = 0.005$), quadriceps muscle strength (SMD = 0.32, 95% CI [0.04 to 0.61], $P = 0.030$), and cost (SMD = -0.50, 95% CI [-0.88 to -0.12], $P = 0.009$).

Ozden et al. [19] analyzed the effects of mobile application-based telerehabilitation, which demonstrated better scores on KOOS (SMD = 0.57, 95% CI [-1.02 to 0.11], $P$ = 0.010).

Liu et al. [20] found a telerehabilitation group exhibited superior outcomes compared to a traditional face-to-face rehabilitation group in terms of extension range (MD = 0.30, 95% CI [0.20 to 0.40], $P$ < 0.001), timed up-and-go test scores (MD = -5.17, 95% CI [-9.79 to -0.55], $P$ = 0.030), VAS scores (MD = -0.43, 95% CI [-0.85 to -0.01], $P$ = 0.040), and KOOS scores (MD = -1.10, 95% CI [-1.63 to 0.57], $P$ < 0.001).

Jiang et al. [40] found telerehabilitation treatment significantly improved WOMAC (MD = 1.13, 95% CI [0.23 to 2.02], $P$ = 0.014), ROM (active extension; MD = 0.30, 95% CI [0.20 to 0.40], $P$ < 0.001), and quadriceps muscle strength (MD = 2.82, 95% CI [1.31 to 4.32], $P$ = 0.001).

Shukia et al. [41] found no significant difference in change in ROM; however, Tsang et al. [42] found telerehabilitation treatment significantly improved ROM (active extension; SMD = -0.19, 95% CI [-0.34 to -0.04], $P$ =0.010).

There were six meta-analyses based on SMDs that showed a significant effect of telerehabilitation interventions on patients undergoing TKA (SMD = 0.03, 95% CI [-0.03 to 0.09], $P$ < 0.001, $I^2$ = 77.0%). (Figure 2). Given the inaccurate CCA, high heterogeneity, and inconsistent assessment efficacy, excluding studies with a high risk of bias and small sample sizes, 53 component RCTs that conducted a sensitivity analysis based on SMDs showed a significant effect of telerehabilitation interventions on patients undergoing TKA (SMD = -0.06, 95% CI [-0.16 to 0.03], $P$ < 0.001, $I^2$ = 69.9%). The bias estimate was -0.4226 (SE = 0.8086, $P$ = 0.604; Figure 3).

Through subgroup analysis, mobile device-assisted, equipment-assisted, and telerehabilitation interventions all showed a significant effect on patients who underwent TKA (SMD = -0.14, 95% CI [-0.28 to 0.01], $P$ < 0.001, $I^2$ = 73.9%; SMD = 0.02, 95% CI [-0.09 to 0.14], $P$ = 0.005, $I^2$ = 47.7%; and SMD = -0.15, 95% CI [-8.53 to 8.23], $P$ < 0.001, $I^2$ = 95.9%, respectively). (Figure 4). Outcomes such as ROM (passive flexion; SMD = 0.26, 95% CI [-0.06 to 0.58], $P$ = 0.041, $I^2$ = 59.8%), WOMAC

(SMD = -0.40, 95% CI [-0.75 to -0.05], $P$ =0.003, $I^2$ = 69.9%), and KOOS (SMD = -0.57, 95% CI [-3.52 to 2.39], $P$ < 0.001, $I^2$ = 85%) all showed improvement over traditional face-to-face rehabilitation among patients who underwent TKA, but there were no statistically significant differences in VAS and ROM (active flexion and extension) (Figure 5).

## Discussion

### Principal findings

To our knowledge, this study presents the first umbrella review exclusively dedicated to telerehabilitation following total knee arthroplasty (TKA), offering a comprehensive and rigorous synthesis of current evidence. By including 14 high-quality systematic reviews and meta-analyses encompassing nearly 200 primary studies, it constitutes the most extensive evaluation of telerehabilitation interventions in this population to date. Through rigorous deduplication and standardization of outcome definitions, we minimized overlap among included reviews and reduced inter-group heterogeneity, thereby enhancing the reliability of the synthesized findings. Our analysis consistently demonstrated that telerehabilitation proved to be more effective and non-inferior, as measured by ROM (passive flexion), WOMAC, and KOOS, than traditional face-to-face rehabilitation among patients who underwent TKA. These benefits were further validated through subgroup analyses across different delivery models (e.g., mobile devices, equipment-assisted, and video conferencing), suggesting a consistent pattern of effects and their potential applicability across various. According to the AMSTAR-2 assessment, half of the included studies provided moderate or higher quality evidence, and some lower-quality studies were primarily limited by the lack of protocol registration and insufficient descriptions of study designs. Additionally, GRADE assessments indicated that factors such as small sample sizes and the absence of blinding contributed to imprecision and a higher risk of bias, leading to downgraded certainty in the evidence for certain outcomes. Despite these limitations, the generally consistent direction and magnitude of favorable effects across multiple domains and review types suggest potential benefits of telerehabilitation. Collectively, the findings indicate that telerehabilitation may serve not only as a supportive adjunct to traditional postoperative care but also as a potentially scalable and accessible approach to improving sustained recovery, particularly for patients facing geographic, mobility, or resource-related challenges.

**Comparison with prior literature**

This study's findings were compared with those of previous studies. Our findings align with the American Physical Therapy Association's clinical guidelines [43] and telemedicine implementation frameworks [44], supporting telerehabilitation (e.g., mobile devices, equipment devices, and video conferencing) as an effective tool for post-TKA rehabilitation. However, heterogeneity across studies was driven by overlapping technologies-such as the combined use of wearable sensors and smartphone apps-making it difficult to isolate individual component effects and limiting subgroup precision. Outcome variability may thus reflect differences in technological integration rather than intervention type. Future studies should adopt standardized reporting (e.g., CONSORT-EHEALTH) to improve comparability. While emerging technologies (e.g., sensors, wearable devices, and chatbots) showed promise in monitoring mobility, ROM, and gait and providing real-time feedback [45,46]. However, variability in gait assessment methods, parameter definitions, and influencing factors across studies hampered the comparability of results. This methodological heterogeneity limited the availability of high-quality randomized controlled trials (RCTs) and robust quantitative meta-analyses. Standardized measurement protocols and consensus on key outcome parameters were urgently needed to strengthen the evidence base and enable more reliable synthesis of findings in future research.

Our umbrella review showed that telerehabilitation significantly increased ROM (passive flexion), WOMAC, and KOOS among patients who underwent TKA; this finding is supported by evidence from both observational studies and RCTs. It also aligns with Christiansen et al. [47], but not Suso-Martí et al. [48]. The reason for the latter was that the assessed cardio-respiratory and musculoskeletal telerehabilitation presented greater variability in the included conditions, which could have influenced the results. However, our umbrella review and the former only focused on patients undergoing TKA, and there was no heterogeneity in the study population. Furthermore, Wei et al.[49] indicated that the above results were crucial monitoring indicators to verify the rehabilitation of TKA patients, and we all confirmed telerehabilitation has a positive effect on these important outcomes. Grassi et al. [50] suggested that passive ROM performed by a surgeon allows a reliable assessment of knee kinematics; active flexion and active extension ROM were radiographically assessed at maximum flexion, whereas passive flexion ROM was clinically assessed with a goniometer [51]. This could explain why in our umbrella review, no significant differences were found regarding active flexion and extension. However, heterogeneity persisted, which may stem from clinical, methodological, and intervention differences—including surgical protocols; follow-up durations and tools; delivery modes; and combined technologies (e.g., apps + wearables). Small samples and lack of blinding in primary studies may have also contributed. Subgroup analyses improved clarity, but standardized designs and reporting are needed to reduce variability.

Our umbrella review found telerehabilitation had no significant effect on pain intensity after TKA, consistent with Kaczorowski et al. [52] and Østerås et al. [53]. This may reflect a shift from acute to chronic pain during extended follow-up, influenced by factors such as age, BMI, diabetes, preoperative pain, and postoperative complications [54]—none directly modifiable by telerehabilitation. In contrast, Tang et al. [55] reported positive effects, likely due to the inclusion of virtual reality (VR) interventions, which have shown efficacy in acute pain relief [21,56–57]. However, we excluded most VR studies conducted during hospitalization according to our criteria. Future research should perform intervention-specific subgroup analyses and conduct high-quality RCTs on VR-based telerehabilitation for post-TKA pain management.

Telerehabilitation had clinically meaningful effects on quality of life, cost, satisfaction, adherence, and adverse events among patients undergoing TKA. These findings are in accordance with those from Lee et al. [43] and Simmich et al. [58]. Some systematic reviews did not conduct statistical comparisons between groups because of different evaluation criteria and questionnaires, but they emphasized that the outcomes of telerehabilitation were not worse than those of conventional face-to-face rehabilitation. Therefore, research should unify the results for each outcome by using a large data model to verify the effectiveness and reliability of telerehabilitation on patients undergoing TKA.

**Implications and future research**

Telerehabilitation represents a novel alternative or complementary therapy that overcomes barriers associated with traditional face-to-face interventions, including reduced travel time, adherence with social distancing measures, and increased accessibility and feasibility [59]. Clinicians, particularly nurses, should consider it as a complementary approach to traditional face-to-face rehabilitation, especially for patients who have limited mobility, geographic barriers, or access challenges. Individualized implementation should account for factors such as age, digital literacy, comorbidities, data security, and reimbursement. These findings also provide actionable insights for policymakers to develop equitable and sustainable telerehabilitation programs. Future studies should adopt large, multi-center RCTs with standardized outcomes (e.g., six-month KOOS, ROM) and adhere to CONSORT guidelines. Clear reporting of telerehabilitation components-technology, durations, and therapist involvement-is critical to reduce heterogeneity. Predictive models should identify responsive patient subgroups to enable precision rehabilitation. Moreover, emerging evidence suggests that a hybrid care- combining initial face-to-face rehabilitation with telerehabiliatation- may maximize both safety and convenience [43]; therefore, future research could explore the optimal combination of face-to-face rehabilitation, telerehabilitation, and their integration, providing important evidence for achieving the best patient outcomes.

**Strengths and Potential Limitations**

This study employed a systematic, transparent methodology following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses reporting checklist (PRISMA 2020), with a prospectively registered protocol in the International Prospective Register of Systematic Reviews (PROSPERO) and use of evidence evaluation tools (AMSTAR 2 and GRADE) to ensure rigor and minimize bias. It uniquely categorized telerehabilitation into mobile, device, and video-assisted programs, offering clarity on telerehabilitation trends. The inclusion of diverse outcomes-functional performance, patient-reported measures, and economic indicators-enhanced clinical and policy relevance. Findings supported the integration of effective telerehabilitation into standard care, particularly for patients who have limited access to in-person services, addressing growing healthcare demands for scalable, cost-effective post-TKA solutions.

This umbrella review is methodologically rigorous, but has limitations, including high heterogeneity of included studies, uneven evidence quality, absence of durable outcome data, insufficient economic analysis, inadequate discussion on patient adherence, and complications and shortcomings with the use of telerehabilitation. Future research should focus on standardized intervention protocols, extended monitoring, cost-effectiveness analysis, and adaptability in diverse populations to enhance the practicality and generalizability of the findings, especially for those who have cognitive impairment, limited digital literacy, or complex comorbidities requiring close supervision.

**Conclusion**

This umbrella review synthesizes the best available evidence from multiple systematic reviews and meta-analyses to evaluate the effectiveness of telerehabilitation following TKA. The findings suggest that telerehabilitation is effective and non-inferior to traditional face-to-face rehabilitation, increasing ROM (passive flexion), WOMAC, KOOS, quality of life, cost, satisfaction, adherence, and adverse events among patients who underwent TKA. Its flexibility and accessibility may help overcome geographical and logistical barriers, thereby enhancing persistent engagement and continuity of care. However, heterogeneity in intervention design, technology platforms, and outcome measures across studies underscores the need for standardized protocols and rigorous methodology in future research. Overall, this review supports the integration of telerehabilitation into post-TKA care pathways, particularly in the context of increasing demand for scalable and patient-centered rehabilitation models. Future high-quality randomized trials, along with hybrid care models, are warranted to confirm these findings and to explore cost-effectiveness, longitudinal outcomes, and equitable access.

**Towards an Integrated Model of Post-TKA Rehabilitation**

Our findings suggest that telerehabilitation is a viable and largely non-inferior alternative to conventional rehabilitation after TKA. However, rather than viewing it as a complete arthroplasty, we advocate for its integration into a broader, patient-centered care pathway. A phased, hybrid model- beginning with face-to-face rehabilitation and transitioning to telerehabilitation- may optimize outcomes by balancing safety, personalization, and accessibility [60]. Future guidelines should define clear indications for each delivery mode and invest in infrastructure to support equitable implementation.

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**Appendix:** Appendix 1-7 .

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**Table1: Characteristics of included systematic reviews.**

| First author, year | Country | Total reviews included | Number of RCTs | Population(s) | | | Interventions | | | Outcome(s) measured | Key results | Meta-analysis | AMS-TAR 2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Sample size(n) | Eligibility criteria | Exclusion criteria | setting | Categorization of the intervention | Duration of the intervention | | | | |
| Liu, 2024[32] | China | 25 | 25 | 4402 | (P) post-TKA patients; (I)use of TELE; (C)traditional FTF rehabilitation; (O)postoperative subjective and objective clinical outcomes, economic indicators, and readmission rates;(S) randomized controlled trials. | Non-RCTs, on-English, inaccessible full-text articles, papers with only abstracts, studies with non-extractable data.involving partial knee replacement patients or those in which patients received both telerehabilitation and faceto-face rehabilitation. | Home | Portable mobile devices (smartphones, smartwatches and tablets for rehabilitation etc.), personal computers for video conferencing through websites, the use of fixed equipment like augmented reality (AR) and virtual reality (VR) devices for TELE. | 2 weeks to 4 months | ROM, PROMs, muscle strength, return to sexual activity, and postoperative readmission rates,quality assessment, pain,gait,Cost, satisfaction, | The TELE group exhibited superior outcomes compared to the traditional face-to-face (FTF) rehabilitation group in terms of pain (standardized mean diferences [SMD]: − 0.15, 95% CI − 0.27 to − 0.04, P = 0.01), passive fexion (MD: 2.60, 95% CI 0.77 to 4.44, P = 0.005), quadriceps muscle strength (SMD: 0.32, 95% CI 0.04 to 0.61, P = 0.03), and cost (SMD: − 0.50, 95% CI − 0.88 to − 0.12, P = 0.009). The subgroup analysis also demonstrated that the fxed equipment-assisted telerehabilitation (FEAT) group and the mobile device-assisted telerehabilitation (MDAT) group were superior to the FTF group. Moreover, patients in the FEAT group exhibited better prognoses than those in the MDAT group. | Yes | Critically low |
| Liu J, 2019[20] | China | 6 | 6 | 601 | (1)RCT.(2)patients who have undergone total knee arthroplasty (TKA). (3)The intervention group receives remote rehabilitation treatment, while the control group receives routine outpatient physical therapy or home visits. (4)Outcome indicators include WOMAC, joint flexion degree, | (1)Data cannot be extracted or the literature is incomplete. (2)Literature is duplicated publication. (3)Non-randomized controlled experiments, reviews, and case studies, etc. (4)Cannot obtain the | home | Video discs, remote video, interactive virtual platforms | Follow- up 6 to 24weeks | WOMAC, ROM, VAS, KOOS, TST, TUGT, muscle strength,SF-36 | There was difference in extension range (MD = 0.30, 95%CI: 0.20 to 0.40, P < 0.001), TUGT scores (MD = -5.17, 95%CI: -9.79 to -0.55, P = 0.03), VAS scores (MD = -0.43, 95%CI: -0.85 to -0.01, P = 0.04) and KOOS scores (MD = -1.10, 95%CI: -1.63 to 0.57, P < 0.0001) between two groups. There was no significant difference in WOMAC scores (MD = -0.32, 95%CI: -2.30 to 1.65, P = 0.75), flexion range (MD = 0.68, 95%CI: -2.28 to 3.63, P = 0.65), and muscle strength (MD = 13.77, 95%CI: -3.89 to | Yes | Critically low |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | VAS, KOOS, TST, TUGT, SF-36, joint range of motion, and muscle strength. | full text | | | | | 31.43, P = 0.13) (P > 0.05) between two groups | | |
| Ozden,F, 2023[19] | Turkey | 584 | 6 | Uncertain | (1) RCTs<br>(2) Post-operative knee arthroplasty cases<br>(3) Mobile application based education, care, rehabilitation or combinations. | (1) Quasi-experimental and pilot studies<br>(2) Conservative and other invasive interventions within the scope of rehabilitation<br>(3) Only abstracts<br>(4) Knee and hip arthroplasty cases. | Family | Mobile application,genus-port device and app;multimedia sources; smartphone and smartwatch; the app was used in combination with the Apple Watch | Intervention duration varied from 4 weeks to 6 weeks and follow-up time was ranging from 4 weeks to 2 years. | ROM,VAS,Psychologi-cal,Satisfaction, compliance,TUG, 10 MWT,30s-CST, 30S-SLS,MUA, KOOS,KSS, and EQ-5D-3L | PEDro scores ranged from 4 to 7 (median:5.5).Mobile application demonstrated better scores on pain, range of motion (ROM), objective and subjective function, satisfaction,compliance in general,<br>and subjective function (ES:0.57, 95% CI: 0.11–1.02) | Yes | Critically low |
| Hameed, 2024[33] | USA | 58 | 22 | 2119 | (1) Used commercially available smartphone applications or wearable activity monitors to measure physical activity or monitor patient health status following arthroplasty procedures; and (2) Reported on relevant outcomes such as patient-reported outcomes, physical activity levels, pain control measures, goal settings, and healthcare utilization | They were published before 2010, were not focused on TKA patients, did not evaluate the effectiveness of commercially available smartphone applications or wearable activity monitors, or were reviews or meta-analyses | family | (1) smartphone applications, (2) wearable devices, and (3) their combination. | Uncertain | Adherence and patient satisfaction, functional outcomes and pain scores, gait analyses and ranges of motion, and measurement and comparison tools | Smartphone applications used for postoperative recovery in TKA showed the potential for improved patient's satisfaction, gait recovery, pain medication scheduling guidance with improved pain management, cost savings, and functional outcomes. Wearable technologies used in postoperative recovery also showed effectiveness in gait and motion analysis. Other demonstrated benefits of the wearables were improved outcomes, return to function, cost reduction, and again, better management of pain due to patient interaction and guidance. Studies that combined applications and wearables demonstrated the individual findings with the addition of adherence, patient ' s satisfaction, and overall mobility improvement at 3 months | No | Critically low |
| Constantinescu,2022[34] | USA | 15 | 6 | 1607 | Commercially available wearable technology or smartphone apps capable of monitoring physical activity and patient status after total knee replacement. | Non-wearable technology,not commercially available devices or apps, pediatric patients, preoperative-only interventions, ongoing | Home and clinical | Smartphone apps, wearable devieces and both | Follow- up 1weeks ~12months | Device accuracy, Pain control, Patient compliance, Physical activity promotion, step count,Healthcare utilization | Commercially available smartphone apps and wearable devices were shown to capably monitor physical activity and improve patient engagement following TKA, making them potentially viable adjuncts or replacements to traditional rehabilitation programs. Components of interventions such as step goals, app-based patient engagement platforms, and | No | Critically low |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | studies, and non-clinical studies such as editorials | | | | | patient-speciffc benchmarks for recovery may improve effectiveness | | |
| Gianzina, 2023[35] | Greece | 12 | uncertain | 474 | (1)Only acceleration sensors for pre- post-operation (2)Sensors for patients at any time point pre- and post-total or partial knee arthroplasty (3)Total or partial knee arthroplasty and gait analysis (4)At least six patients Studies published after 1998 (5)Studies comparing the outcome between patients who have knee osteoarthritis | Book chapters Review papers Conference papers Article abstracts or unavailable as full texts Insufficient data for analysis Results not related to our inclusion criteria Did not measure unassisted gait at a self selected speed | Not mentioned | Wearable sensors | 5days-1years;follow-up 6weeks,3months, 1year | WOMAC,KSS,OKS, EuroQoL ,VAS and other clinical outcome scores; walking speed, cadence, step irregularity and other Gait variables | Acceleration-based gait analysis has notable clinical use in monitoring patients after TKA. This application provides objective information on the functional outcome beyond the use of clinical outcome scores | No | Critically low |
| Gordon , 2025[36] | USA | 31 | 31 | 4666 | (1) Utilized commercially available smartphone applications or wearable activity monitors to assess physical activity or monitor patient health status following TKA procedures; (2) Reported relevant outcomes such as patient-reported outcomes, physical activity levels, pain management, goal attainment, and healthcare utilization. | Studies published before 2010 that were not focused on TKA patients or those that did not assess the effectiveness of commercially available smartphone applications or wearable activity monitors were excluded. | Family | 1)smartphone applications, 2) wearable devices, and 3) their combined use. | 1~12months | Patient satisfaction scores and adherences, functional outcomes, and pain scores, ranges of motion and gait analyses, and measurements and comparison tools | Smartphone applications highlighted benefits, including patient satisfaction, improved gait, optimized pain management through medication scheduling guidance, cost savings, and better functional outcomes; wearable technologies have advantages on monitoring and effectiveness in gait and motion analysis and additional benefits included improved recovery outcomes, enhanced return to function, cost reduction, and better pain management through patient interaction and guidance; integrated both applications and wearables, corroborated with patient satisfaction and overall mobility enhancement at three months post surgery | No | Critically low |
| King, 2023[37] | UK | 14 | 11 | 142 | their interventions included the use of wearable sensor technology in the outpatient setting following primary TKA or unicompartmental knee | Simultaneous or sequential bilateral knee surgery was included while revision TKA was excluded | Family | wearable devices (joint-specific or physical activity sensors) | joint-specific: varied from 10 days to 3 months physical activity sensors: varied | Weekly therapy visits,TUG test, unilateral balance test, stair climb test or 6 MW distance, | Remotely supervised rehabilitation using wearable sensors demonstrated similar outcomes when provided as an alternative to standard care in most studies. One group found improved outcomes for knee-specific sensors compared with standard care. | No | Critically low |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | arthroplasty (UKA). | | | | from 2 weeks to 6 months | ADLS,KOOS, NPRS,VR-12,knee ROM,hamstring strength and VAS, Quadriceps strength,MVPA, healthcare costs, resource use | There were improved physical activity and healthcare resource use outcomes described in the literature where sensors were used in addition to standard care | | |
| Pang,2023[38] | China | 13 | 13 | 230~6418 | Not mentioned | Not mentioned | Family | tele-rehabilitation, video-conferencing, virtual reality rehabilitation, internet rehabilitation, robot-assisted rehabilitation | Not mentioned | Pain,WOMAC, KOOS,knee flexion and extension, walking ability and quality of life, Adverse events, satisfaction, cost | Telerehabilitation positively affects walking ability, knee extension and patient costs post-TKA surgery. Regarding the quality of life, patient satisfaction and the WOMAC, telerehabilitation had similar effects to conventional rehabilitation. | No | Critically low |
| Salehian, 2025[39] | Iran | 18 | 9 | Uncertain | (1)Only considered full-text journal articles (2)English language (3)Were published during 2019–2023 (4)Involving mobile health applications that provide telerehabilitation are implicated. | (1)Abstracts,review studies,editorials, conferences,brief reports,and dissertations (2)Not relevant to the research question | Not mentioned | Sensors, wearable activity trackers, artificial intelligence, Chatbots, smartwatch, the Internet of Things, and Machine Learning | Not mentioned | KOOS,PROM, EQ-5D-5L,WOMAC,SF-12 | Telerehabilitation can be as effective as traditional rehabilitation in improving the condition of patients after TKA | No | Critically low |
| Feng,2023 [22] | China | 25 | Uncertain | 823 | (1) gait analyses with wearable sensors; (2) post-TKA management; and (3) studies published between January 2010 and March 2023 | (1) conference abstracts; (2) review articles; (3) non-TKA treatment; (4) the lack of gait or biomechanical data; (5) studies on technological evaluation; (6) the lack of wearable sensors; (7) robot-assisted rehabilitation or the use of a surgical navigation | Not mentioned | Wearable sensors | 1weeks to 5years Follow- up 6weeks,3months, 6months, and 1year after surgery | PROMS and gait variables | Gait analysis using wearable sensors and PROMs showed diferences in controlled environments, daily life, and when comparing diferent surgeries | No | Critically low |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | system; (8) non-independent walking; and (9) the absence of full text. | | | | | | | |
| Jiang,2018 [40] | China | 4 | 4 | 422 | Study population, patients after TKA;intervention, tele-rehabilitation;control, face-to-face approaches; outcome measures, the change in visual analogue scale (VAS) scores or WOMAC change; and study design, RCT. | Uncertain | Home | Places equipment in patient's homes, remote supervised rehabilitation used communication technologies | Follow- up 6~24 weeks | VAS scores, WOMAC change, active flexion range, extension range, and quadriceps strength,adverse events | Telerehabilitation could achieve comparable pain relief (mean differences ¼ 0.52; 95% confidence intervals (CIs) ¼ 0.20 to 1.24; p ¼ 0.16) and better Western Ontario and McMaster Universities Osteoarthritis Index improvement (mean differences ¼ 1.13; 95% CI ¼ 0.23 to 2.02; p ¼ 0.014). In addition, telerehabilitation treatment resulted in a significantly higher extension range (p < 0.00001) and quadriceps strength (p ¼ 0.0002) than face-to-face rehabilitation | Yes | Critically low |
| Shukla, 2017[41] | India | 6 | 6 | 408 | Patients following TKA, ability to walk, having at least 90°active knee flexion range of motion prior to TKA13 and post TKA active range of motion: flexion 80° and extension10° | Patients who have sensory, cognitive, and/or praxic impairment, and concomitant medical conditions that may influence the rehabilitation process. | Family | Telephone-based method of telerehabilitation, videoconferencing connected to patients' homes via internet, interactive virtual telerehabilitation knee rehabilitation software | 2~12weeks study and 3months~40 weeks follow- up | Knee extension and flexion; quadriceps muscle strength; The timed up-and-go test; and assessment of pain, stiffness, and functional capacity using the WOMAC and VAS | There was no significant difference in change in active knee extension and flexion in the home telerehabilitation group as compared to the control group (mean differences (MD) 0.52, 95% CI 1.39 to 0.35, p ¼ 0.24 and MD 1.14, 95% CI 0.61 to 2.89, p ¼ 0.20, respectively). | Yes | Critically low |
| Tsang,2024 [42] | China | 11 | 11 | 1825 | (P)Patients were discharged after TKR. (I)Used tele-rehabilitation , (C)Control group receiving traditional in-person rehabilitation (O)Postoperative clinical, patient-reported or financial outcome measures (S)interventional studies In English. | (1)Other languages (2) Population includes other than patients who underwent TKR (3)Abstract-only papers (4)No full text available | Home | Patients who have sensory, cognitive, and/or praxic impairment, and concomitant medical conditions that may influence the | Immediately after the intervention to 52 weeks after surgery | Knee range of motion, muscle strength, pain, WOMAC;KOOS, cost, adverse events | No significant difference was found in the improvement of pain and physical function in patients (Standardized Mean Differences (SMD)–0.15, 95% CI −0.47 to 0.16, P =0.34 and SMD –0.04, 95% CI−0.19 to 0.12, P =0.62, respectively). The utilization of hospital resources and costs were significantly lower in telerehabilitation. | Yes | Critically low |

Table2 Summary of the effectiveness of telerehabilitation after total knee arthroplasty

| Study | Intervention | Outcome | Number of studies | Total participants(EG/CG) | Metrics | Effect size(95%CI) | $I^2$ | *P*-value | GRADE |
|---|---|---|---|---|---|---|---|---|---|
| Liu,2024[32] | A | ROM(Active Flexion) | 5 | 857(426/431) | SMD | 1.18(-0.20,2.57) | 23 | 0.09 | Moderate |
| Liu,2024[32] | B | ROM(Active Flexion) | 6 | 778(388/390) | SMD | 1.73(0.07,3.40) | 4 | 0.04 | Moderate |
| Ozden,2023[19] | A | ROM(Active Flexion) | 2 | 469(183/286) | MD | 0.28(-0.31,-0.79) | 82.8 | 0.39 | Low |
| Liu,2019[20] | C | ROM(Active Flexion) | 4 | 532(267/265) | MD | 0.68(-2.28,3.63) | 86 | 0.65 | Low |
| Jiang,2018[40] | C | ROM(Active Flexion) | 2 | 198(99/99) | MD | 2.40(-0.34,5.15) | 0 | 0.09 | Moderate |
| Shukla,2017[41] | C | ROM(Active Flexion) | 3 | 148(124/124) | MD | 1.14(-0.61,2.89) | 4 | 0.20 | Moderate |
| Tsang,2024[42] | C | ROM(Active Flexion) | 5 | 886(444/442) | SMD | 0.12(-0.07,0.30) | 45 | 0.22 | Moderate |
| Liu,2024[32] | A | ROM(Active Extension) | 3 | 663(326/337) | SMD | -0.27(-0.70,0.15) | 0 | 0.21 | Moderate |
| Liu,2024[32] | B | ROM(Active Extension) | 6 | 778(388/390) | SMD | -0.25(-0.75,0.25) | 0 | 0.32 | Moderate |
| Liu,2019[20] | C | ROM(Active Extension) | 3 | 412(207/205) | MD | 0.30(0.20,0.40) | 0 | <0.0001 | Moderate |
| Jiang,2018[40] | C | ROM(Active Extension) | 3 | 396(197/199) | MD | 0.30(0.20,0.40) | 0 | <0.00001 | Moderate |
| Shukla,2017[41] | C | ROM(Active Extension) | 3 | 248(124/124) | MD | -0.52(-1.39,0.35) | 0 | 0.24 | Moderate |
| Tsang,2024[42] | C | ROM(Active Extension) | 4 | 699(350,349) | SMD | -0.19(-0.34,-0.04) | 0 | 0.01 | Moderate |
| Liu,2024[32] | A | ROM(Passive Flexion) | 2 | 547(261/286) | SMD | 2.60(0.77,4.44) | 42 | 0.005 | Moderate |
| Liu,2024[32] | B | ROM(Passive Flexion) | 3 | 303(150/153) | SMD | 3.27(0.17,6.37) | 38 | 0.04 | Moderate |
| Liu,2024[32] | A | WOMAC | 2 | 220(110/110) | SMD | -1.97(-3.99,0.04) | 57 | 0.05 | Low |
| Liu,2024[32] | B | WOMAC | 4 | 362(179/183) | SMD | -1.02(-1.90,-0.15) | 0 | 0.02 | Low |
| Liu,2019[20] | C | WOMAC | 4 | 532(267/265) | MD | -0.32(-2.30,1.65) | 90 | 0.75 | Low |
| Jiang,2018[40] | C | WOMAC | 2 | 270(135/135) | MD | 1.13(0.23,2.20) | 0 | 0.809 | Moderate |
| Liu,2024[32] | A | VAS | 4 | 570(295/275) | SMD | -0.15(-0.27,-0.04) | 0 | 0.010 | Moderate |
| Liu,2024[32] | B | VAS | 5 | 568(283/285) | SMD | -0.13(-0.29,0.04) | 0 | 0.13 | Moderate |
| Liu,2019[20] | C | VAS | 3 | 236(117/119) | MD | -0.43(-0.84,-0.01) | 0 | 0.04 | Low |
| Jiang,2018[40] | C | VAS | 2 | 161(81/80) | MD | 0.52(-0.20,1.24) | 0 | 0.16 | Moderate |
| Shukla,2017[41] | C | VAS | 2 | 225(111/114) | MD | -0.02(-0.46,0.41) | 0 | 0.91 | Moderate |
| Tsang,2024[42] | C | VAS | 4 | 681(340/341) | SMD | -0.15(-0.64,0.34) | 74 | 0.34 | Moderate |
| Ozden,2023[19] | A | KOOS | 2 | 550(267/283) | SMD | 0.23(0.11,-1.02) | 85.1 | 0.01 | Low |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Liu,2019[20] | C | KOOS | 2 | 234(116/118) | MD | -1.10(-1.63,-0.57) | 0 | <0.0001 | Low |
| Liu,2019[20] | C | TUGT | 3 | 248(124/124) | MD | -5.17(-9.79,-0.55) | 84 | 0.03 | Very Low |
| Liu,2019[20] | C | Muscle strength | 2 | 207(103/104) | MD | 13.77(-3.89,31.43) | 99 | 0.001 | Very Low |
| Jiang,2018[40] | C | Muscle strength | 2 | 198(99/99) | MD | 2.82(1.31,4.32) | 0 | 0.0002 | Moderate |
| Tsang,2024[42] | C | Stiffness | 2 | 255(128/127) | SMD | 0.23(-0.44,0.91) | 83 | 0.50 | Very Low |
| Tsang,2024[42] | C | Function | 3 | 642(319/323) | SMD | -0.04(-0.19,0.12) | 0 | 0.62 | Moderate |

A:MDAT(Mobile Device-Assisted Telerehabilitation)B:FEAT(Fixed Equipment-Assisted Telerehabilitation) C:Telerehabilitation(Both).

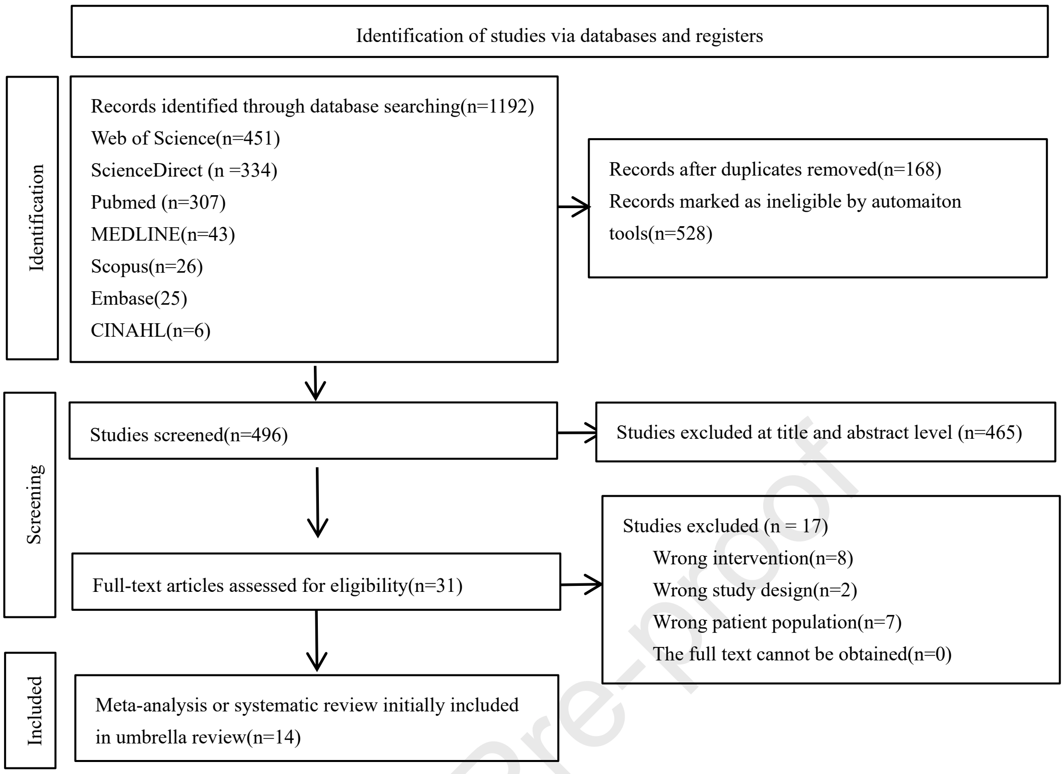



Figure1:Flowchart of the literature search and screening process

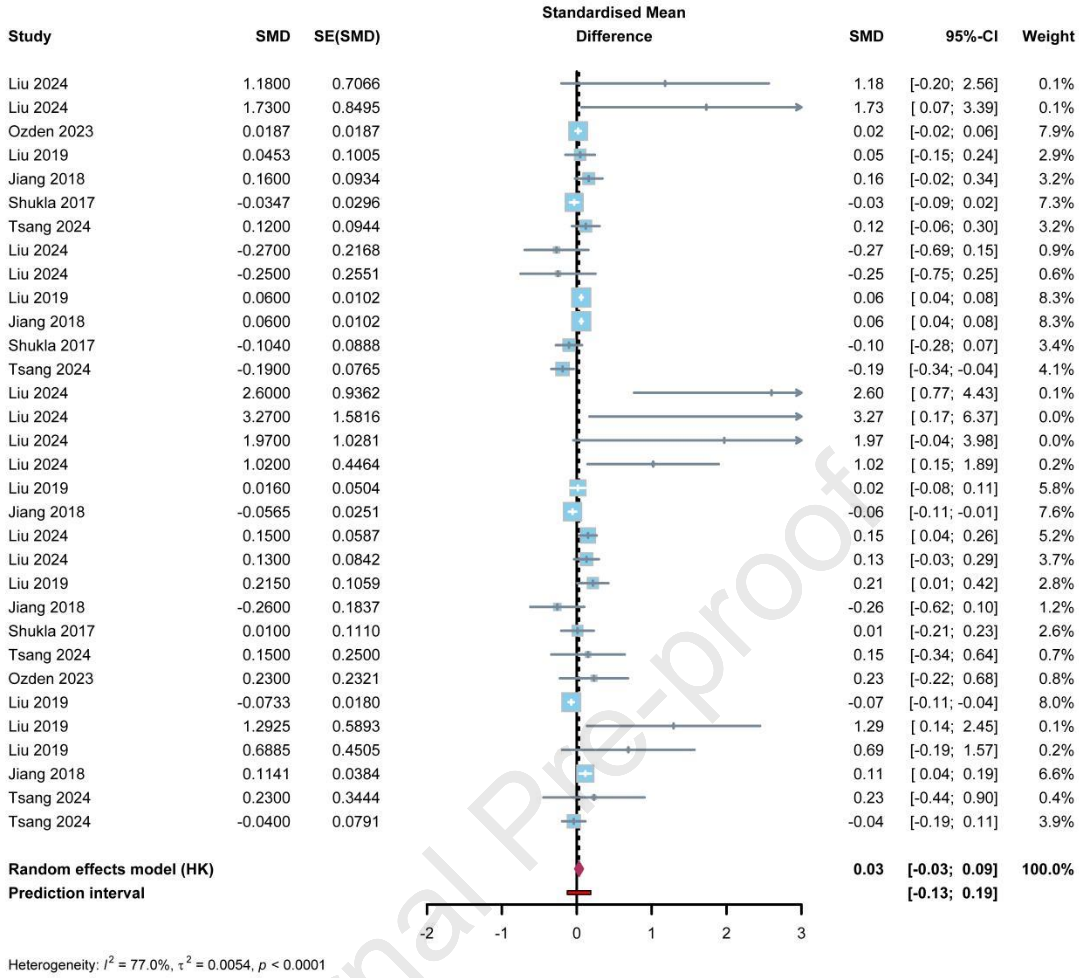
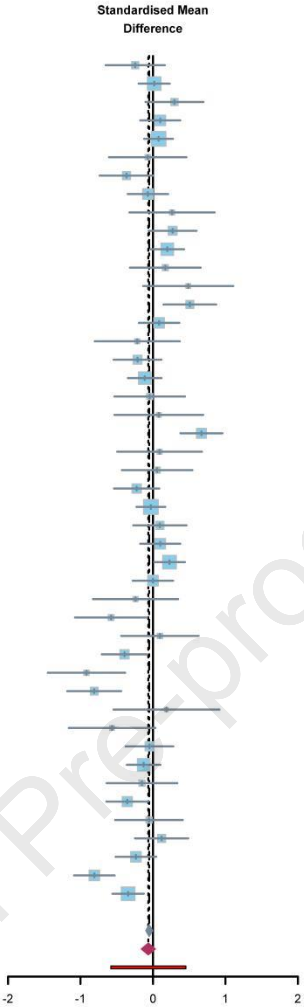



Figure2: Forest plots of main analysis of Meta-analysis. A negative value represents an effect in favor of telerehabilitation. Effect sizes and 95% confidence intervals from meta-analyses are represented by a diamond. Bold lines of equivalence represent borders of clinical meaningfulness [SMD]=0.03, 95% CI= -0.03 to 0.09, *p*<0.0001 and $I^2$=77.0%.




| | Intervention group | | | Control group | | | | | Weight | Weight |
|---|---|---|---|---|---|---|---|---|---|---|
| **Study** | **Total** | **Mean** | **SD** | **Total** | **Mean** | **SD** | **SMD** | **95%-CI** | **(common)** | **(random)** |
| Huang,et al-2017 | 30 | -2.15 | 5.8400 | 102 | -0.83 | 5.2100 | -0.24 | [-0.65; 0.16] | 1.4% | 2.0% |
| Tripuraneni,et al-2021 | 153 | -11.70 | 17.6000 | 184 | -12.00 | 18.3000 | 0.02 | [-0.20; 0.23] | 4.9% | 2.9% |
| Campbell,et al-2019 | 48 | 111.90 | 13.3000 | 49 | 108.00 | 12.8000 | 0.30 | [-0.10; 0.70] | 1.4% | 2.1% |
| Chen,et al-2016 | 101 | 86.70 | 9.6000 | 101 | 85.70 | 10.2000 | 0.10 | [-0.18; 0.38] | 3.0% | 2.6% |
| Han,et al-2015 | 194 | 96.80 | 13.3000 | 196 | 95.70 | 15.1000 | 0.08 | [-0.12; 0.28] | 5.8% | 3.0% |
| Kramer,et al-2003 | 27 | 117.90 | 11.8000 | 27 | 118.80 | 12.8000 | -0.07 | [-0.61; 0.46] | 0.8% | 1.6% |
| Pua,et al-2023 | 56 | 103.00 | 12.0000 | 58 | 108.00 | 15.0000 | -0.36 | [-0.74; 0.01] | 1.7% | 2.2% |
| Moffet,et al-2015 | 98 | 108.90 | 9.9000 | 100 | 109.60 | 10.0000 | -0.07 | [-0.35; 0.21] | 2.9% | 2.6% |
| Nuevo,et al-2023 | 23 | 105.22 | 8.4600 | 22 | 102.27 | 13.1600 | 0.26 | [-0.32; 0.85] | 0.7% | 1.4% |
| Piqueras,et al-2013 | 72 | 18.16 | 9.7100 | 70 | 15.63 | 8.8200 | 0.27 | [-0.06; 0.60] | 2.1% | 2.4% |
| Prvu,et al-2020 | 143 | 114.50 | 15.3000 | 144 | 111.40 | 16.0000 | 0.20 | [-0.03; 0.43] | 4.2% | 2.8% |
| Russel,et al-2011 | 31 | 19.82 | 10.7800 | 34 | 17.82 | 12.3100 | 0.17 | [-0.32; 0.66] | 1.0% | 1.7% |
| Tousignant,et al-2011 | 21 | 110.79 | 11.5500 | 20 | 104.57 | 13.5100 | 0.49 | [-0.14; 1.11] | 0.6% | 1.3% |
| Sun,et al-2017 | 60 | 93.50 | 7.6000 | 60 | 89.40 | 8.4000 | 0.51 | [ 0.14; 0.87] | 1.7% | 2.2% |
| Moffet,et al-2015 | 98 | 4.80 | 4.0000 | 100 | 4.50 | 3.0000 | 0.08 | [-0.19; 0.36] | 2.9% | 2.6% |
| Nuevo,et al-2023 | 23 | 5.00 | 4.7700 | 22 | 6.14 | 5.5500 | -0.22 | [-0.80; 0.37] | 0.7% | 1.4% |
| Piqueras,et al-2013 | 72 | 0.20 | 2.8000 | 70 | 0.90 | 3.7000 | -0.21 | [-0.54; 0.12] | 2.1% | 2.4% |
| Prvu,et al-2020 | 143 | 2.50 | 3.6000 | 144 | 2.90 | 3.4000 | -0.11 | [-0.35; 0.12] | 4.2% | 2.8% |
| Russel,et al-2011 | 31 | 3.45 | 3.4500 | 34 | 3.62 | 3.9400 | -0.05 | [-0.53; 0.44] | 1.0% | 1.7% |
| Tousignant,et al-2011 | 21 | 2.92 | 3.5700 | 20 | 2.61 | 3.9600 | 0.08 | [-0.53; 0.69] | 0.6% | 1.4% |
| Fleischman,et al-2018 | 96 | 125.04 | 7.4700 | 97 | 120.00 | 7.5400 | 0.67 | [ 0.38; 0.96] | 2.7% | 2.6% |
| Nuevo,et al-2023 | 23 | 106.10 | 8.8000 | 22 | 105.20 | 10.6000 | 0.09 | [-0.49; 0.68] | 0.7% | 1.4% |
| Russel,et al-2011 | 31 | 17.89 | 10.5000 | 34 | 17.17 | 13.8600 | 0.06 | [-0.43; 0.54] | 1.0% | 1.7% |
| Campbell,et al-2019 | 76 | 1.60 | 2.8000 | 83 | 2.60 | 5.5000 | -0.23 | [-0.54; 0.09] | 2.3% | 2.5% |
| Han,et al-2015 | 194 | 6.30 | 7.3000 | 196 | 6.50 | 6.3000 | -0.03 | [-0.23; 0.17] | 5.8% | 3.0% |
| Pua,et al-2023 | 56 | 3.20 | 4.6000 | 58 | 2.80 | 3.7000 | 0.10 | [-0.27; 0.46] | 1.7% | 2.2% |
| Chen,et al-2016 | 101 | 92.20 | 10.2000 | 101 | 91.20 | 9.7000 | 0.10 | [-0.18; 0.38] | 3.0% | 2.6% |
| Crawford,et al-2021 | 160 | 121.20 | 8.9000 | 185 | 118.80 | 11.7000 | 0.23 | [ 0.02; 0.44] | 5.0% | 2.9% |
| Moffet,etal-2015 | 98 | 78.50 | 12.9000 | 100 | 78.50 | 13.0000 | 0.00 | [-0.28; 0.28] | 2.9% | 2.6% |
| Nuevo,et al-2023 | 23 | 18.49 | 11.5100 | 22 | 21.34 | 11.7600 | -0.24 | [-0.83; 0.35] | 0.7% | 1.4% |
| Russel,et al-2011 | 31 | -3.26 | 2.0400 | 34 | -2.16 | 1.7200 | -0.58 | [-1.08; -0.08] | 0.9% | 1.7% |
| Shim,et al-2023 | 27 | 46.50 | 9.1000 | 27 | 45.30 | 15.0000 | 0.10 | [-0.44; 0.63] | 0.8% | 1.6% |
| Kramer,et al-2003 | 80 | 57.23 | 35.9600 | 80 | 71.29 | 35.3100 | -0.39 | [-0.71; -0.08] | 2.3% | 2.5% |
| Sahin,et al-2022 | 30 | 27.97 | 4.6700 | 30 | 31.57 | 2.8600 | -0.92 | [-1.45; -0.38] | 0.8% | 1.6% |
| Sun,et al-2017 | 60 | 23.20 | 8.7000 | 60 | 29.50 | 6.6000 | -0.81 | [-1.18; -0.44] | 1.6% | 2.2% |
| Bini,et al-2017 | 14 | -3.43 | 2.6800 | 15 | -4.00 | 3.2300 | 0.19 | [-0.54; 0.92] | 0.4% | 1.1% |
| Nuevo,et al-2023 | 23 | 1.55 | 1.2600 | 22 | 2.55 | 2.1300 | -0.56 | [-1.16; 0.03] | 0.6% | 1.4% |
| Piqueras,et al-2013 | 72 | -0.69 | 1.4400 | 70 | -0.61 | 1.8700 | -0.05 | [-0.38; 0.28] | 2.1% | 2.4% |
| Prvu,et al-2020 | 143 | 2.70 | 2.0000 | 144 | 3.00 | 2.6000 | -0.13 | [-0.36; 0.10] | 4.2% | 2.8% |
| Russel,et al-2011 | 31 | 3.07 | 1.5500 | 34 | 3.29 | 1.3100 | -0.15 | [-0.64; 0.34] | 1.0% | 1.7% |
| Chen,et al-2016 | 94 | 1.61 | 0.6200 | 93 | 1.84 | 0.6700 | -0.35 | [-0.64; -0.07] | 2.7% | 2.6% |
| Pronk,et al-2020 | 38 | 1.40 | 1.6000 | 33 | 1.50 | 2.0000 | -0.06 | [-0.52; 0.41] | 1.0% | 1.8% |
| Pua,et al-2023 | 56 | 2.00 | 1.9000 | 58 | 1.80 | 1.4000 | 0.12 | [-0.25; 0.49] | 1.7% | 2.2% |
| Timmers,et al-2019 | 107 | 3.99 | 4.6200 | 91 | 5.08 | 4.6200 | -0.24 | [-0.52; 0.05] | 2.9% | 2.6% |
| Timmers,et al-2019 | 114 | -8.30 | 9.2000 | 99 | -0.75 | 9.4200 | -0.81 | [-1.09; -0.53] | 2.9% | 2.6% |
| Tripuraneni,et al-2021 | 153 | 12.10 | 9.9500 | 184 | 15.60 | 10.3500 | -0.34 | [-0.56; -0.13] | 4.9% | 2.9% |
| **Common effect model** | **3376** | | | **3529** | | | **-0.05** | **[-0.10; -0.00]** | **100.0%** | . |
| **Random effects model** | | | | | | | **-0.06** | **[-0.16; 0.03]** | . | **100.0%** |
| **Prediction interval** | | | | | | | | **[-0.58; 0.45]** | | |

Heterogeneity: $I^2 = 69.9\%$, $\tau^2 = 0.0637$, $p < 0.0001$

Figure3: Forest plots of Sensitivity analysis of RCTs. A negative value represents an effect in favor of telerehabilitation. Effect sizes and 95% confidence intervals from meta-analyses are represented by a diamond. Bold lines of equivalence represent borders of clinical meaningfulness [SMD]= -0.06,95%CI = -0.16 to 0.03, $p$<0.0001 and $I^2$=69.9%. Bias estimate:-0.4226(SE=0.8086), $p$=0.604.

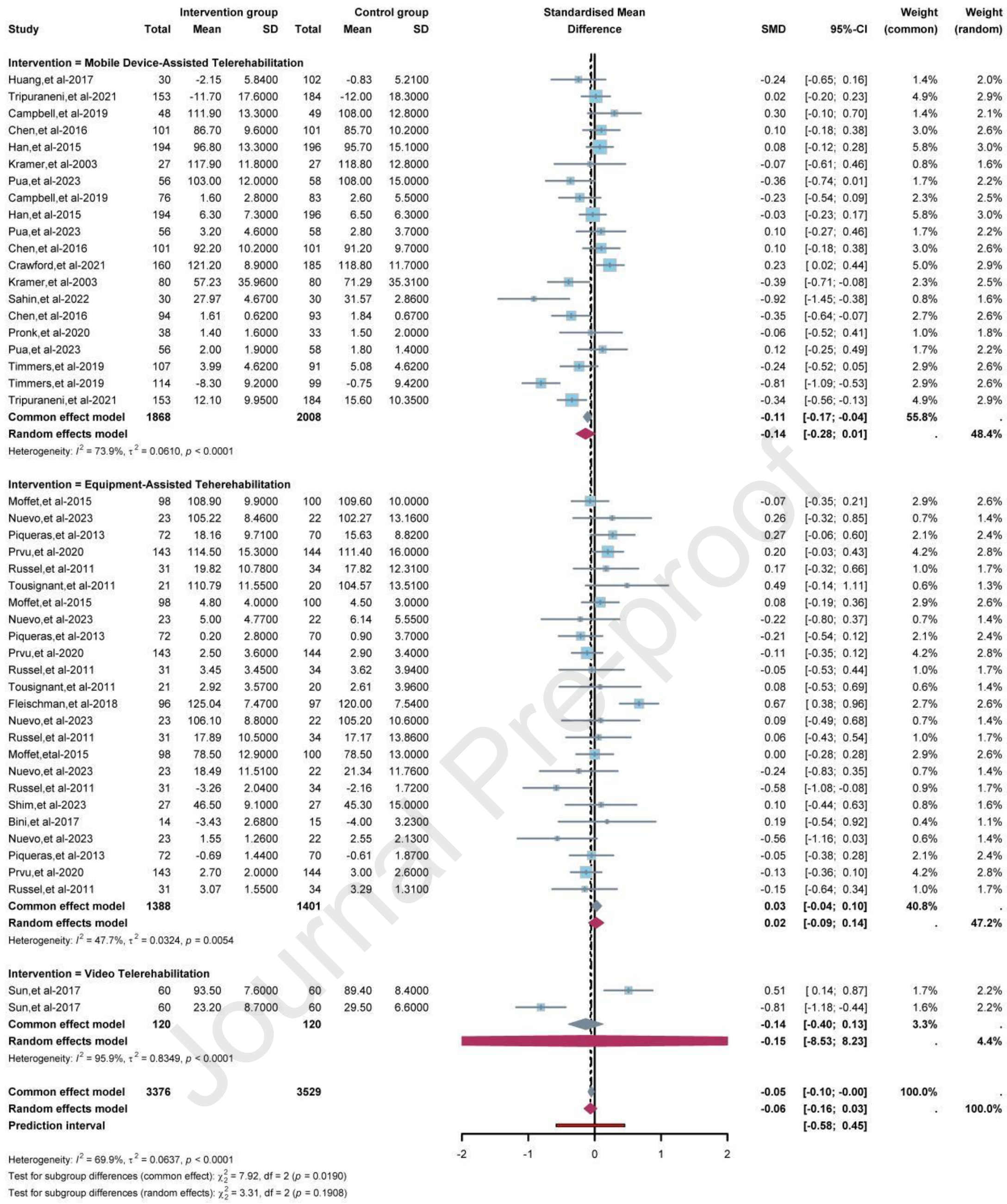
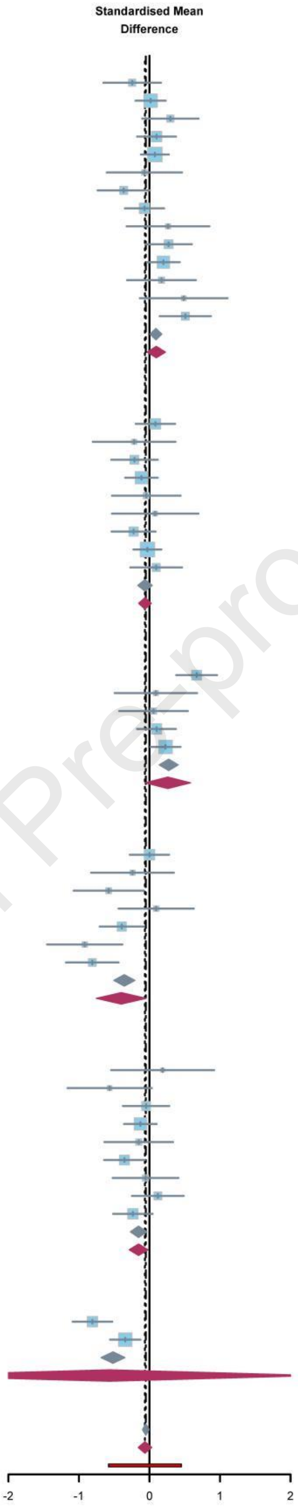



Figure4: Forest plots of intervation. A:MDAT(Mobile Device-Assisted Telerehabilitation); B:FEAT(Fixed Equipment-Assisted Telerehabilitation); C: Video Telerehabilitation(Both).




| Study | Intervention group Total | Intervention group Mean | Intervention group SD | Control group Total | Control group Mean | Control group SD | SMD | 95%-CI | Weight (common) | Weight (random) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Outcome = ROM(Active fexion)** | | | | | | | | | | |
| Huang,et al-2017 | 30 | -2.15 | 5.8400 | 102 | -0.83 | 5.2100 | -0.24 | [-0.65; 0.16] | 1.4% | 2.0% |
| Tripuraneni,et al-2021 | 153 | -11.70 | 17.6000 | 184 | -12.00 | 18.3000 | 0.02 | [-0.20; 0.23] | 4.9% | 2.9% |
| Campbell,et al-2019 | 48 | 111.90 | 13.3000 | 49 | 108.00 | 12.8000 | 0.30 | [-0.10; 0.70] | 1.4% | 2.1% |
| Chen,et al-2016 | 101 | 86.70 | 9.6000 | 101 | 85.70 | 10.2000 | 0.10 | [-0.18; 0.38] | 3.0% | 2.6% |
| Han,et al-2015 | 194 | 96.80 | 13.3000 | 196 | 95.70 | 15.1000 | 0.08 | [-0.12; 0.28] | 5.8% | 3.0% |
| Kramer,et al-2003 | 27 | 117.90 | 11.8000 | 27 | 118.80 | 12.8000 | -0.07 | [-0.61; 0.46] | 0.8% | 1.6% |
| Pua,et al-2023 | 56 | 103.00 | 12.0000 | 58 | 108.00 | 15.0000 | -0.36 | [-0.74; 0.01] | 1.7% | 2.2% |
| Moffet,et al-2015 | 98 | 108.90 | 9.9000 | 100 | 109.60 | 10.0000 | -0.07 | [-0.35; 0.21] | 2.9% | 2.6% |
| Nuevo,et al-2023 | 23 | 105.22 | 8.4600 | 22 | 102.27 | 13.1600 | 0.26 | [-0.32; 0.85] | 0.7% | 1.4% |
| Piqueras,et al-2013 | 72 | 18.16 | 9.7100 | 70 | 15.63 | 8.8200 | 0.27 | [-0.06; 0.60] | 2.1% | 2.4% |
| Prvu,et al-2020 | 143 | 114.50 | 15.3000 | 144 | 111.40 | 16.0000 | 0.20 | [-0.03; 0.43] | 4.2% | 2.8% |
| Russel,et al-2011 | 31 | 19.82 | 10.7800 | 34 | 17.82 | 12.3100 | 0.17 | [-0.32; 0.66] | 1.0% | 1.7% |
| Tousignant,et al-2011 | 21 | 110.79 | 11.5500 | 20 | 104.57 | 13.5100 | 0.49 | [-0.14; 1.11] | 0.6% | 1.3% |
| Sun,et al-2017 | 60 | 93.50 | 7.6000 | 60 | 89.40 | 8.4000 | 0.51 | [ 0.14; 0.87] | 1.7% | 2.2% |
| **Common effect model** | **1057** | | | **1167** | | | **0.09** | **[ 0.01; 0.18]** | **32.0%** | . |
| **Random effects model** | | | | | | | **0.10** | **[-0.03; 0.23]** | . | **31.0%** |
| Heterogeneity: $I^2 = 36.8\%$, $\tau^2 = 0.0156$, $p = 0.0821$ | | | | | | | | | | |
| **Outcome = ROM(Active Extension)** | | | | | | | | | | |
| Moffet,et al-2015 | 98 | 4.80 | 4.0000 | 100 | 4.50 | 3.0000 | 0.08 | [-0.19; 0.36] | 2.9% | 2.6% |
| Nuevo,et al-2023 | 23 | 5.00 | 4.7700 | 22 | 6.14 | 5.5500 | -0.22 | [-0.80; 0.37] | 0.7% | 1.4% |
| Piqueras,et al-2013 | 72 | 0.20 | 2.8000 | 70 | 0.90 | 3.7000 | -0.21 | [-0.54; 0.12] | 2.1% | 2.4% |
| Prvu,et al-2020 | 143 | 2.50 | 3.6000 | 144 | 2.90 | 3.4000 | -0.11 | [-0.35; 0.12] | 4.2% | 2.8% |
| Russel,et al-2011 | 31 | 3.45 | 3.4500 | 34 | 3.62 | 3.9400 | -0.05 | [-0.53; 0.44] | 1.0% | 1.7% |
| Tousignant,et al-2011 | 21 | 2.92 | 3.5700 | 20 | 2.61 | 3.9600 | 0.08 | [-0.53; 0.69] | 0.6% | 1.4% |
| Campbell,et al-2019 | 76 | 1.60 | 2.8000 | 83 | 2.60 | 5.5000 | -0.23 | [-0.54; 0.09] | 2.3% | 2.5% |
| Han,et al-2015 | 194 | 6.30 | 7.3000 | 196 | 6.50 | 6.3000 | -0.03 | [-0.23; 0.17] | 5.8% | 3.0% |
| Pua,et al-2023 | 56 | 3.20 | 4.6000 | 58 | 2.80 | 3.7000 | 0.10 | [-0.27; 0.46] | 1.7% | 2.2% |
| **Common effect model** | **714** | | | **727** | | | **-0.06** | **[-0.17; 0.04]** | **21.2%** | . |
| **Random effects model** | | | | | | | **-0.06** | **[-0.15; 0.03]** | . | **20.0%** |
| Heterogeneity: $I^2 = 0\%$, $\tau^2 = 0$, $p = 0.8198$ | | | | | | | | | | |
| **Outcome = ROM(Passive fexion)** | | | | | | | | | | |
| Fleischman,et al-2018 | 96 | 125.04 | 7.4700 | 97 | 120.00 | 7.5400 | 0.67 | [ 0.38; 0.96] | 2.7% | 2.6% |
| Nuevo,et al-2023 | 23 | 106.10 | 8.8000 | 22 | 105.20 | 10.6000 | 0.09 | [-0.49; 0.68] | 0.7% | 1.4% |
| Russel,et al-2011 | 31 | 17.89 | 10.5000 | 34 | 17.17 | 13.8600 | 0.06 | [-0.43; 0.54] | 1.0% | 1.7% |
| Chen,et al-2016 | 101 | 92.20 | 10.2000 | 101 | 91.20 | 9.7000 | 0.10 | [-0.18; 0.38] | 3.0% | 2.6% |
| Crawford,et al-2021 | 160 | 121.20 | 8.9000 | 185 | 118.80 | 11.7000 | 0.23 | [ 0.02; 0.44] | 5.0% | 2.9% |
| **Common effect model** | **411** | | | **439** | | | **0.27** | **[ 0.14; 0.41]** | **12.3%** | . |
| **Random effects model** | | | | | | | **0.26** | **[-0.06; 0.58]** | . | **11.3%** |
| Heterogeneity: $I^2 = 59.8\%$, $\tau^2 = 0.0397$, $p = 0.0411$ | | | | | | | | | | |
| **Outcome = WOMAC** | | | | | | | | | | |
| Moffet,etal-2015 | 98 | 78.50 | 12.9000 | 100 | 78.50 | 13.0000 | 0.00 | [-0.28; 0.28] | 2.9% | 2.6% |
| Nuevo,et al-2023 | 23 | 18.49 | 11.5100 | 22 | 21.34 | 11.7600 | -0.24 | [-0.83; 0.35] | 0.7% | 1.4% |
| Russel,et al-2011 | 31 | -3.26 | 2.0400 | 34 | -2.16 | 1.7200 | -0.58 | [-1.08; -0.08] | 0.9% | 1.7% |
| Shim,et al-2023 | 27 | 46.50 | 9.1000 | 27 | 45.30 | 15.0000 | 0.10 | [-0.44; 0.63] | 0.8% | 1.6% |
| Kramer,et al-2003 | 80 | 57.23 | 35.9600 | 80 | 71.29 | 35.3100 | -0.39 | [-0.71; -0.08] | 2.3% | 2.5% |
| Sahin,et al-2022 | 30 | 27.97 | 4.6700 | 30 | 31.57 | 2.8600 | -0.92 | [-1.45; -0.38] | 0.8% | 1.6% |
| Sun,et al-2017 | 60 | 23.20 | 8.7000 | 60 | 29.50 | 6.6000 | -0.81 | [-1.18; -0.44] | 1.6% | 2.2% |
| **Common effect model** | **349** | | | **353** | | | **-0.36** | **[-0.51; -0.21]** | **10.0%** | . |
| **Random effects model** | | | | | | | **-0.40** | **[-0.75; -0.05]** | . | **13.6%** |
| Heterogeneity: $I^2 = 69.9\%$, $\tau^2 = 0.1013$, $p = 0.0028$ | | | | | | | | | | |
| **Outcome = VAS** | | | | | | | | | | |
| Bini,et al-2017 | 14 | -3.43 | 2.6800 | 15 | -4.00 | 3.2300 | 0.19 | [-0.54; 0.92] | 0.4% | 1.1% |
| Nuevo,et al-2023 | 23 | 1.55 | 1.2600 | 22 | 2.55 | 2.1300 | -0.56 | [-1.16; 0.03] | 0.6% | 1.4% |
| Piqueras,et al-2013 | 72 | -0.69 | 1.4400 | 70 | -0.61 | 1.8700 | -0.05 | [-0.38; 0.28] | 2.1% | 2.4% |
| Prvu,et al-2020 | 143 | 2.70 | 2.0000 | 144 | 3.00 | 2.6000 | -0.13 | [-0.36; 0.10] | 4.2% | 2.8% |
| Russel,et al-2011 | 31 | 3.07 | 1.5500 | 34 | 3.29 | 1.3100 | -0.15 | [-0.64; 0.34] | 1.0% | 1.7% |
| Chen,et al-2016 | 94 | 1.61 | 0.6200 | 93 | 1.84 | 0.6700 | -0.35 | [-0.64; -0.07] | 2.7% | 2.6% |
| Pronk,et al-2020 | 38 | 1.40 | 1.6000 | 33 | 1.50 | 2.0000 | -0.06 | [-0.52; 0.41] | 1.0% | 1.8% |
| Pua,et al-2023 | 56 | 2.00 | 1.9000 | 58 | 1.80 | 1.4000 | 0.12 | [-0.25; 0.49] | 1.7% | 2.2% |
| Timmers,et al-2019 | 107 | 3.99 | 4.6200 | 91 | 5.08 | 4.6200 | -0.24 | [-0.52; 0.05] | 2.9% | 2.6% |
| **Common effect model** | **578** | | | **560** | | | **-0.15** | **[-0.27; -0.04]** | **16.7%** | . |
| **Random effects model** | | | | | | | **-0.15** | **[-0.29; -0.02]** | . | **18.7%** |
| Heterogeneity: $I^2 = 0\%$, $\tau^2 = 0$, $p = 0.4764$ | | | | | | | | | | |
| **Outcome = KOOS** | | | | | | | | | | |
| Timmers,et al-2019 | 114 | -8.30 | 9.2000 | 99 | -0.75 | 9.4200 | -0.81 | [-1.09; -0.53] | 2.9% | 2.6% |
| Tripuraneni,et al-2021 | 153 | 12.10 | 9.9500 | 184 | 15.60 | 10.3500 | -0.34 | [-0.56; -0.13] | 4.9% | 2.9% |
| **Common effect model** | **267** | | | **283** | | | **-0.52** | **[-0.69; -0.35]** | **7.7%** | . |
| **Random effects model** | | | | | | | **-0.57** | **[-3.52; 2.39]** | . | **5.5%** |
| Heterogeneity: $I^2 = 85\%$, $\tau^2 = 0.0920$, $p = 0.0099$ | | | | | | | | | | |
| **Common effect model** | **3376** | | | **3529** | | | **-0.05** | **[-0.10; -0.00]** | **100.0%** | . |
| **Random effects model** | | | | | | | **-0.06** | **[-0.16; 0.03]** | . | **100.0%** |
| **Prediction interval** | | | | | | | | **[-0.58; 0.45]** | | |

Heterogeneity: $I^2 = 69.9\%$, $\tau^2 = 0.0637$, $p < 0.0001$
Test for subgroup differences (common effect): $\chi^2_5 = 80.33$, df = 5 ($p < 0.0001$)
Test for subgroup differences (random effects): $\chi^2_5 = 27.26$, df = 5 ($p < 0.0001$)

Figure5: Forest plots of outcomes.Abbreviation:ROM:Range of motion; WOMAC:the Western Ontario and McMaster Universities Osteoarthritis Index;VAS: Visual analogue scale; KOOS:Knee Injury and Osteoarthritis Outcome Score.